Image Manipulations using OpenCV

ARTI 404 - Image Processing | Lab 3

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Objective: Apply geometric and intensity transformations to a color image and compare the resulting images.



Input image as displayed in the executed Jupyter notebook.

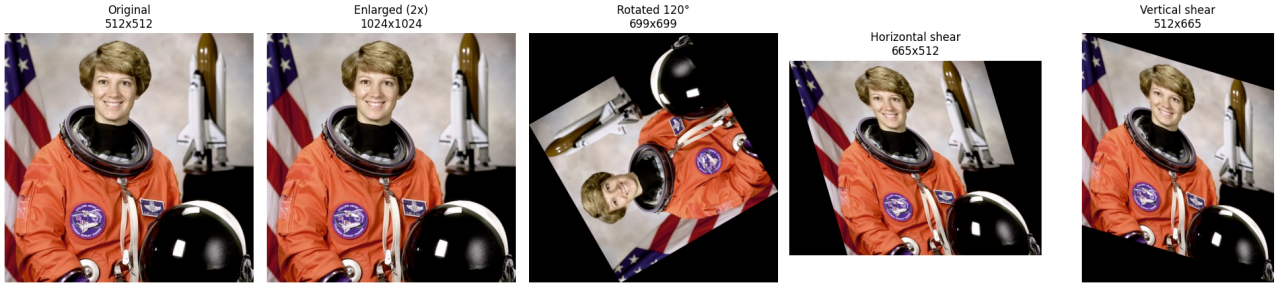
Tools: Python, OpenCV, NumPy, and Matplotlib. The recorded input dimensions are 512 x 512 pixels. OpenCV reads the image in BGR order; the notebook converts it to RGB for display.

Task 1 - Geometric transformations

Enlargement: The image is scaled by a factor of 2 in both directions using cv2.resize with INTER_CUBIC interpolation. Its dimensions change from 512 x 512 to 1024 x 1024.

Rotation: cv2.getRotationMatrix2D generates a rotation of 120 degrees around the image center. The translation is adjusted and the canvas is expanded to 699 x 699 to retain the rotated image. Black pixels fill regions outside the source image.

Shear: Horizontal and vertical affine shear use a coefficient of 0.3. Horizontal shear applies x' = x + 0.3y, producing 665 x 512 pixels. Vertical shear applies y' = y + 0.3x, producing 512 x 665 pixels.



Recorded notebook results: original, enlarged, rotated, horizontal shear, and vertical shear. Dimensions are width x height.

Observation: Enlargement increases the pixel dimensions. Rotation changes orientation, while shear slants the image by shifting rows or columns proportionally to their position.

Task 2 - Intensity transformations

Negative: s = 255 - r is applied to each color channel. Bright values become dark and dark values become bright, producing complementary colors.

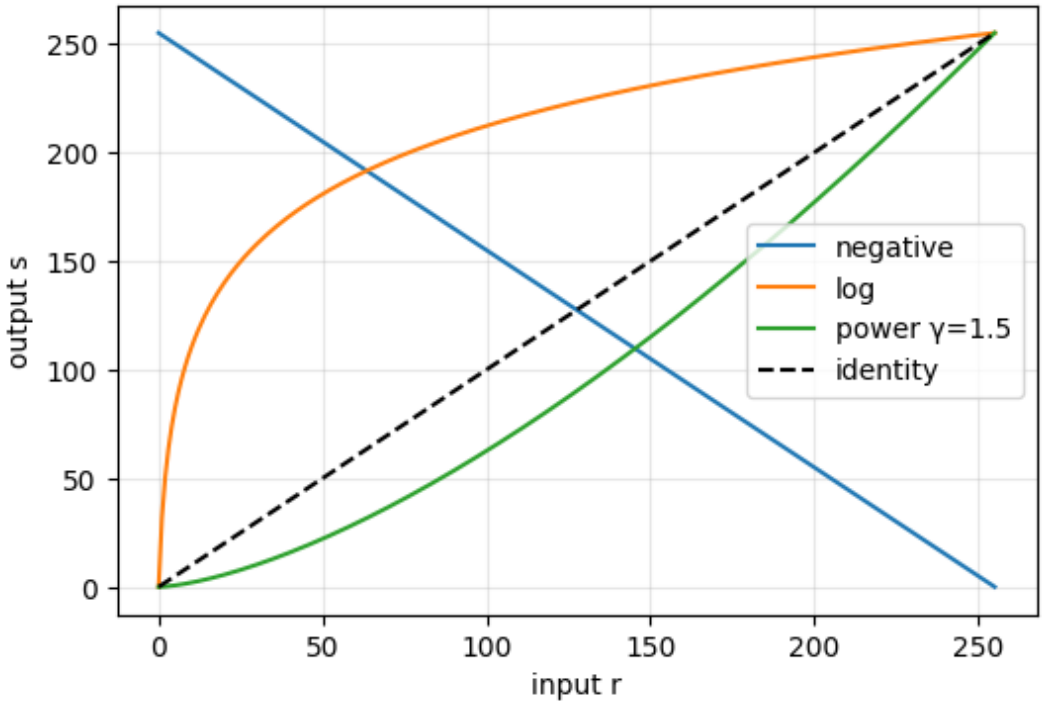
Log transformation: s = c ln(1 + r), with c = 255 / ln(256), approximately 45.99. The input is converted to floating point before calculation, and the output is clipped and converted to uint8. Dark intensities are expanded, so shadows become brighter.

Power-law transformation: s = 255 $(r / 255)^{gamma}$, with gamma = 1.5. This value darkens intermediate intensities and expands differences at the bright end. The result shows stronger separation in some bright regions; improvement in contrast depends on the image and intensity range.



Recorded notebook results: original, negative, log, and power-law (gamma = 1.5).

Transformation curves and conclusion



The dashed identity line represents unchanged intensities. The log curve lies above it, gamma = 1.5 lies below it, and the negative curve reverses the intensity scale.

Conclusion: All requested operations are implemented in the notebook: enlargement, 120-degree rotation, shear, negative, logarithmic, and power-law transformations. Geometric operations change spatial arrangement, while intensity operations change pixel values.

Submission contents: The Lab3 folder contains the Jupyter notebook, this report, the recovered input image, and the recorded output images.

Image provenance: Figures in this report are taken from the saved outputs of the supplied notebook. The original input file was absent from the supplied package; images/input.jpg is reconstructed from the displayed image and resized to the recorded 512 x 512 dimensions. It has lower detail than the original source. Individual result PNGs are extracted previews; the comparison figures preserve the recorded notebook output.